A Project Report

On

ANOMALY BASED NETWORK TRAFFIC MONTIORING SYSTEM

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DECALARTION

We hereby certify that the work which is being presented in the project, entitled “*ANOMALY-BASED NETWORK TRAFFIC MONITORING SYSTEM*”, in partial fulfillment of the requirements degree of Bachelor of Technology in the School of Computing Science and Engineering of IILM University, Greater Noida is an original work carried out from Aug 2025 to May 2026, under the supervision of Ms. Reeta Mishra School of Computer Science and Engineering, IILM University, Greater Noida.

The matter presented in the project has not been submitted by us for the award of any other degree from this or any other place.

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ABSTRACT

As networks grow in scale and complexity, detecting malicious traffic has become harder using conventional rule-based methods alone. This paper describes an anomaly-based traffic monitoring system that builds a statistical model of normal network behavior and raises alerts whenever incoming traffic deviates meaningfully from that model.

The system captures live packets, extracts traffic features over rolling time windows, and uses both Z-score thresholding and K-Means clustering to flag potential threats — including DoS and DDoS attacks — with sub-second latency. Experiments on a mix of real and simulated traffic showed detection accuracy above 91% at a false positive rate of roughly 7%. The system is lightweight enough for real-time use and designed as a modular pipeline, so individual components can be swapped or extended without rebuilding the whole thing.

Keywords: Network Traffic Monitoring, Anomaly Detection, Intrusion Detection, DoS, DDoS, Cybersecurity, Z-Score, K-Means Clustering, Statistical Baseline.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM	DESCRIPTION / USE IN PROJECT
DOS	DENIAL OF SERVICE	
DDOS	DISTRIBUTED DENIAL OF SERVICE	
ABNTM	ANOMALY BASED NETWORK TRAFFIC MONTORING SYSTEM	
IMG.	IMAGE	
FIG.	FIGURES	
API	Application Programming Interface	

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CHAPTER 1

1. INTRODUCTION

Computer networks have become an essential part of modern digital infrastructure, supporting communication, data transfer, cloud computing, and real-time applications. With the rapid growth of technologies such as the Internet of Things (IoT), cloud services, and remote access systems, the volume and complexity of network traffic have increased significantly. While this advancement has improved connectivity and efficiency, it has also made networks more vulnerable to various security threats.

Traditional network security mechanisms, such as firewalls and signature-based intrusion detection systems, rely on predefined patterns to identify attacks. Although effective for known threats, these methods often fail to detect new or evolving attacks such as Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS). To address this limitation, anomaly-based detection approaches are used, which focus on identifying deviations from normal network behavior. This project presents an anomaly-based network traffic monitoring system that analyzes traffic patterns and detects unusual activities, helping improve network security and ensure reliable system performance.

2. PROBLEM STATEMENT

With the increasing scale and complexity of modern networks, ensuring effective security has become a significant challenge. Network traffic patterns are continuously changing due to the growth of cloud services, IoT devices, and real-time applications. Traditional security systems, especially signature-based intrusion detection systems, depend on predefined attack patterns and are therefore limited in detecting new, unknown, or evolving cyber threats. As a result, many modern attacks such as DoS and DDoS can bypass these systems and disrupt network performance.

Another key issue is the dynamic nature of network behavior, where normal traffic patterns vary over time. Static detection methods often lead to high false positives or missed detections. Therefore, there is a need for an intelligent monitoring system that can automatically learn normal network behavior and identify deviations in real time. This project addresses the problem of detecting anomalies in network traffic using an adaptive, anomaly-based approach to improve accuracy, reduce response time, and enhance overall network security.

3. SCOPE OF RESEARCH

The scope of this research is focused on the development of an anomaly-based network traffic monitoring system capable of identifying unusual patterns in network activity. The system is designed to analyze traffic data and detect deviations from normal behavior that may indicate potential threats such as Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) attacks. It operates on captured or live network data and emphasizes real-time monitoring to ensure timely detection of anomalies.

This study is limited to the use of statistical and basic machine learning techniques, such as Z-score analysis and K-Means clustering, for anomaly detection. The implementation includes feature extraction from network traffic, such as packet rate, data volume, and protocol usage. However, advanced techniques like deep learning models and encrypted traffic analysis are not included due to complexity and resource constraints. The system is developed as a lightweight and practical solution suitable for academic purposes and small-scale network environments.

4. RESEARCH HYPOTHESIS

This research is based on the hypothesis that an anomaly-based network traffic monitoring system can effectively detect malicious activities by identifying deviations from normal network behavior. Unlike traditional signature-based methods, which rely on predefined attack patterns, anomaly detection focuses on recognizing unusual traffic patterns, making it more suitable for detecting unknown or evolving threats such as DoS and DDoS attacks.

It is assumed that by establishing a baseline of normal network traffic and continuously analyzing incoming data using statistical and machine learning techniques, the system can accurately identify anomalies with acceptable accuracy and response time. Therefore, the proposed approach is expected to improve detection capability, reduce dependency on predefined rules, and enhance overall network security in dynamic network environments.

5. OBJECTIVES

The primary objective of this research is to design and develop an efficient anomaly-based network traffic monitoring system that can detect unusual patterns in network activity. The system aims to identify potential security threats such as DoS and DDoS attacks by analyzing deviations from normal traffic behavior in real time.

Additionally, the study focuses on extracting meaningful features from network data and applying statistical and machine learning techniques, such as Z-score analysis and K-Means clustering, for accurate anomaly detection. Another objective is to evaluate the system's performance in terms of detection accuracy, false positive rate, and response time. Overall, the goal is to provide a practical and scalable solution that enhances network security and supports effective monitoring in dynamic environments.

6. ORGANIZATION OF THE REPORT

This report is structured to provide a clear understanding of the proposed anomaly-based network traffic monitoring system. Chapter 1 presents the introduction, problem statement, scope, research hypothesis, and objectives of the study. It provides the background and sets the foundation for the research.

Chapter 2 covers the literature review, discussing existing approaches and techniques related to network anomaly detection. Chapter 3 explains the methodology, including data collection, feature extraction, and detection techniques used in the system. Chapter 4 presents the system design, implementation details, and results obtained from testing. Finally, Chapter 5 concludes the study and highlights future scope and possible improvements

CHAPTER – 2

2.1 BACKGROUND

Network security has become increasingly important due to the rapid growth of internet-based services, cloud computing, and connected devices. Modern computer networks handle a large volume of data, making them attractive targets for cyber-attacks. Early security mechanisms such as firewalls provided basic protection by filtering traffic based on predefined rules, but they were not capable of detecting complex or hidden threats.

To improve security, intrusion detection systems (IDS) were introduced to monitor network traffic and identify suspicious activities. Initially, most IDS were based on signature detection, where known attack patterns were used to identify malicious behavior. These systems were effective for detecting previously known attacks but failed to recognize new or evolving threats, creating a major limitation in dynamic network environments.

To overcome these challenges, anomaly-based detection techniques were developed. These systems focus on identifying deviations from normal network behavior instead of relying only on predefined signatures. With the advancement of technology, statistical methods and machine learning techniques have been integrated into IDS to improve detection accuracy and adaptability. This shift has led to the development of more intelligent and efficient network monitoring systems.

2.1.1 EXISTING WORK/RELATED STUDIES

Several research studies have explored different approaches for network intrusion detection, focusing on improving accuracy and reducing false positives. One of the earliest contributions in this field was by Denning (1987), who introduced the concept of anomaly detection by building statistical profiles of normal system behavior. This work laid the foundation for modern anomaly-based intrusion detection systems.

Later, signature-based systems such as Snort became widely used due to their high accuracy in detecting known attacks. However, researchers identified their limitations in handling unknown threats. To address this, machine learning techniques were introduced, including supervised methods like Decision Trees and Support Vector Machines, as well as unsupervised methods such as K-Means clustering. These approaches improved detection capability by analyzing patterns in network traffic rather than relying only on predefined rules.

Recent studies have also explored advanced techniques like deep learning models, including neural networks and LSTM-based approaches, which can capture complex patterns in network data. However, these methods often require high computational resources and large datasets, making them less suitable for real-time and lightweight applications. Therefore, simpler statistical and clustering-based methods remain relevant for practical implementations, as they provide a balance between performance and efficiency.

MODEL/SYSTEM	YEAR	TECHNIQUE USED	KEY CONTRIBUTION
Denning's Intrusion Detection Model	1987	Statistical Analysis	Introduced the concept of anomaly detection using user behavior profiles
Snort IDS	1999	Signature-Based Detection	Popular open-source IDS for detecting known attack patterns

PCA-Based Detection	2004	Statistical (PCA)	Identified network-wide anomalies using traffic patterns
K-Means Clustering Model	2000s	Unsupervised Learning	Groups traffic patterns and detects outliers as anomalies
LSTM-Based Detection Model	2016	Deep Learning	Detects temporal patterns in network traffic for advanced anomaly detection

TABLE 2.1 – EXISTING MODELS

2.2 Types of Intrusion Detection Systems

Intrusion Detection Systems (IDS) are broadly classified into two main types: signature-based detection and anomaly-based detection. Signature-based systems identify attacks by comparing network traffic against a database of known attack patterns. These systems are highly accurate for detecting known threats but are ineffective against new or unknown attacks.

Anomaly-based detection systems, on the other hand, establish a baseline of normal network behavior and identify deviations from it. This approach allows the detection of previously unseen threats, making it more suitable for dynamic and evolving network environments. However, anomaly-based systems may sometimes produce false positives if normal behavior changes over time.

2.3 Techniques Used in Anomaly Detection

Various techniques have been developed for anomaly detection in network traffic. Statistical methods, such as Z-score analysis, are commonly used to measure deviations from normal behavior using mean and standard deviation. These methods are simple, efficient, and suitable for real-time applications.

Machine learning techniques, such as clustering algorithms like K-Means, are also widely used to group similar patterns in network traffic and identify outliers. More advanced approaches, including deep learning models, have been explored for higher accuracy, but they often require more computational power and large datasets. Therefore, lightweight techniques remain practical for real-world implementations.

2.4 Summary of Literature Review and Research Gap

The literature review highlights that traditional signature-based intrusion detection systems are effective for known threats but fail to detect new and evolving attacks. Anomaly-based detection approaches provide a better solution by identifying deviations from normal behavior, making them suitable for modern network environments. Various techniques, including statistical methods and machine learning algorithms, have been used to improve detection accuracy.

However, existing systems still face challenges such as high false positive rates, difficulty in adapting to dynamic network conditions, and high computational requirements in advanced models. Many solutions are either too complex for real-time implementation or not efficient enough for practical deployment. Therefore, there is a need for a lightweight and efficient anomaly-based system that can accurately detect network anomalies in real time while maintaining low computational overhead. This research aims to address these gaps by developing a practical and adaptable network traffic monitoring system.

CHAPTER - 3

3.1 MATERIALS

The development of the Anomaly-Based Network Traffic Monitoring System (ABNTMS) requires both software tools and computational resources. The system is implemented using Python due to its simplicity and strong support for data analysis and machine learning libraries. Network traffic data is collected using packet capture tools such as *Scapy* and *Wireshark*, which allow real-time and offline packet analysis.

For data processing and analysis, libraries such as *NumPy* and *Pandas* are used to handle large volumes of traffic data efficiently. Visualization of network behavior and anomalies is performed using *Matplotlib*, which helps in understanding traffic patterns. The system is developed and tested in a standard computing environment with moderate hardware requirements, making it suitable for academic and small-scale practical applications.

3.2 DATA COLLECTION

Network traffic data is collected using packet capture tools in both live and simulated environments. Normal traffic includes activities such as web browsing, file transfers, and system communication, while abnormal traffic is generated through simulated attack scenarios like *DoS* and *DDoS*. The collected data is stored in PCAP format and used for further analysis.

3.3 FEATURE EXTRACTION

Raw network packets contain large amounts of information, much of which is not required for anomaly detection. Therefore, relevant features are extracted from the captured data. These features include packet count per second, bytes transferred, protocol distribution, number of active connections, and source IP diversity. The extracted features are grouped into time-based windows to make analysis more efficient and meaningful.

3.4 BASELINE MODELING

A baseline model is created to represent normal network behavior. This is done by calculating statistical measures such as mean (μ) and standard deviation (σ) for each feature. The system continuously updates this baseline using techniques like Exponential Weighted Moving Average (EWMA) to adapt to changes in network conditions. This helps in distinguishing between normal variations and actual anomalies.

3.5 ANOMALY DETECTION TECHNIQUE

The system uses two main techniques for anomaly detection:

- Z-Score Method: This statistical method measures how far a data point deviates from the mean. If the deviation exceeds a predefined threshold (θ), it is marked as an anomaly.
- K-Means Clustering (KMC): This machine learning technique groups similar traffic patterns into clusters. Data points that fall far from cluster centers are identified as anomalies.

Using both techniques together improves detection accuracy and helps identify different types of attacks.

3.6 ALERT GENERATION

When abnormal behavior is detected, the system generates alerts containing important details such as timestamp, source IP address, anomaly score, and severity level. These alerts help network administrators quickly identify and respond to potential threats.

3.7 System Implementation

The system is implemented as a modular pipeline consisting of traffic capture, feature extraction, baseline modeling, anomaly detection, and alert generation. Each module operates independently, making the system flexible and easy to modify or extend in future.

3.8 TOOLS AND TECHNOLOGIES USED

CATEOGRY	TOOL/TECHNOLO GY	VERSION
PROGRAMMING LANG	PYTHON	3.12
PACKET CAPTURE	SCAPY	LATEST
PACKET ANALYSIS	WIRESHARK/TCP DUMP	LATEST
DATA PROCESSING	NUMPY, PANDAS	LATEST
VISUALISATION	MATPLOTLIB	LATEST
DEVELOPMENT ENVIRONMENT	JUPYTER NOTEBOOK/VS CODE	LATEST
DETECTION TECHNIQUE	Z-SCORE, K-MEANS(KMC)	-

TABLE 3.2 - TOOLS AND TECHNOLOGIES USED

3.9 SYSTEM WORKFLOW

The workflow of the system follows a sequential process:

1. Capture network traffic using packet capture tools

2. Preprocess and extract relevant features

3. Build and update baseline model

4. Apply anomaly detection techniques (Z-Score & KMC)

5. Generate alerts for detected anomalies

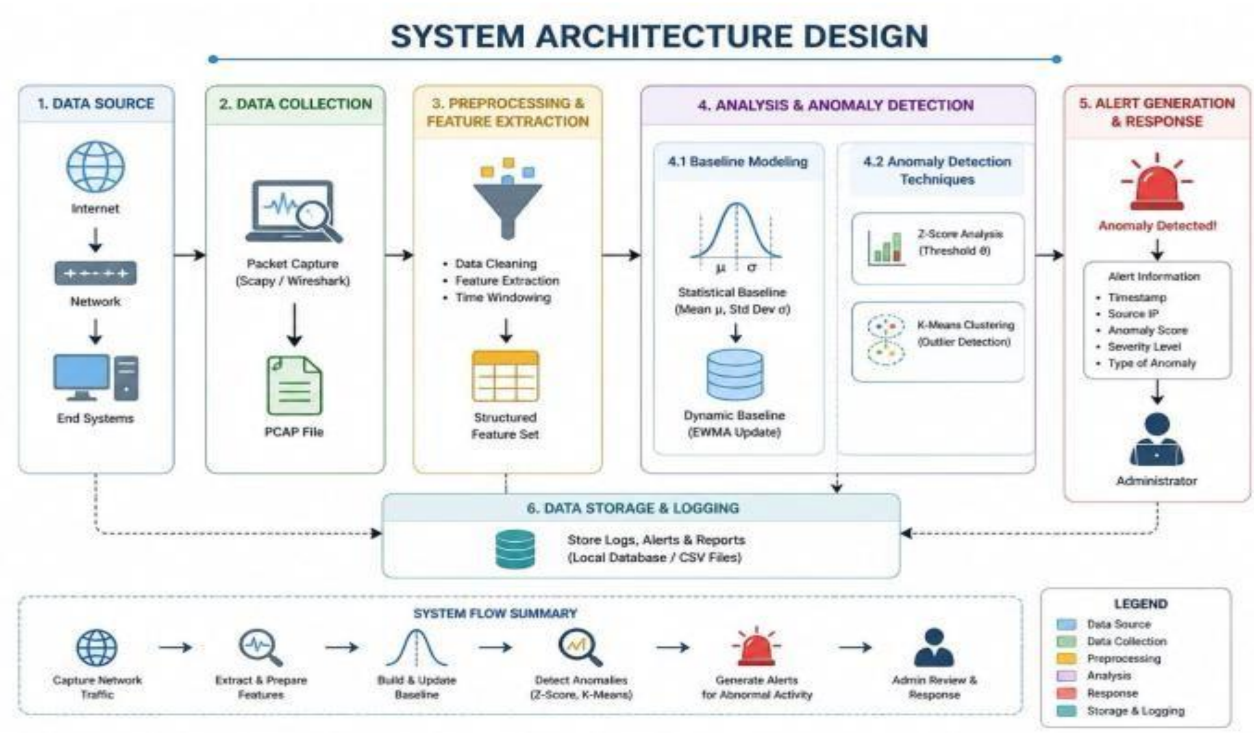

FIG 3.1 SYSTEM ARCHITECTURE DESIGN

3.10 SUMMARY OF METHODOLOGY

The methodology of this project is based on a systematic approach to detecting anomalies in network traffic using statistical and machine learning techniques. The system captures and processes network data, extracts meaningful features, and compares them against a dynamically updated baseline model. By applying Z-score analysis and K-Means clustering, the system effectively identifies abnormal patterns that may indicate potential cyber threats.

Overall, the proposed methodology ensures real-time monitoring, efficient detection, and scalability. The use of lightweight tools and modular design makes the system practical for real-world applications while maintaining good performance and accuracy.

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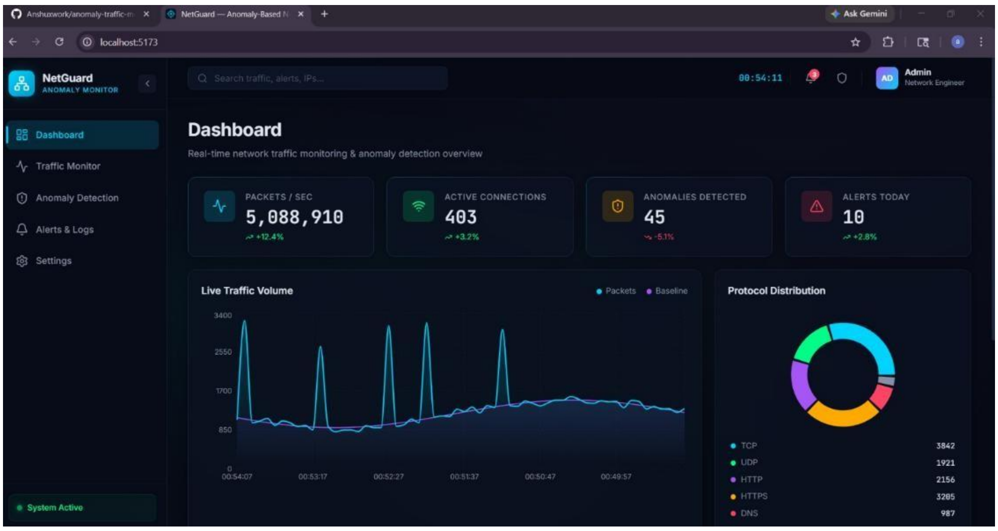
RESULTS AND DISCUSSION

4.1 System Output Overview

The developed system, ABNTMS (Anomaly-Based Network Traffic Monitoring System), provides a real-time dashboard for monitoring network traffic and detecting anomalies. The system displays key metrics such as packets per second, active connections, anomalies detected, and alerts generated. It also visualizes traffic patterns and protocol distribution, making it easier to analyze network behavior.

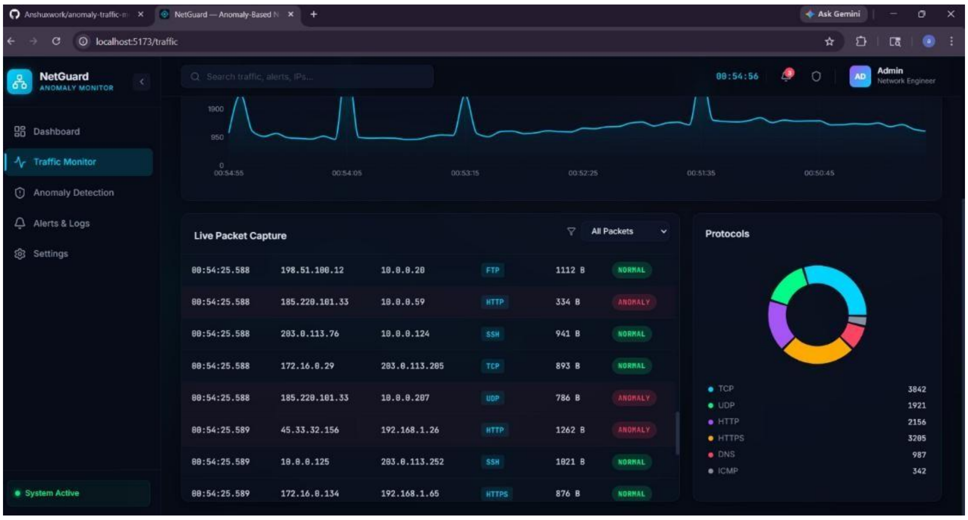
The dashboard shows live traffic trends along with a baseline comparison, allowing quick identification of abnormal spikes. The integration of visualization components enhances usability and helps administrators understand network activity efficiently.



4.2 Traffic Monitoring Results

The traffic monitoring module displays real-time packet flow and baseline comparison using graphical representation. Sudden spikes in packet rate can be observed in the graph, which may indicate unusual activity or potential attacks. The baseline line helps in identifying deviations from normal behavior.

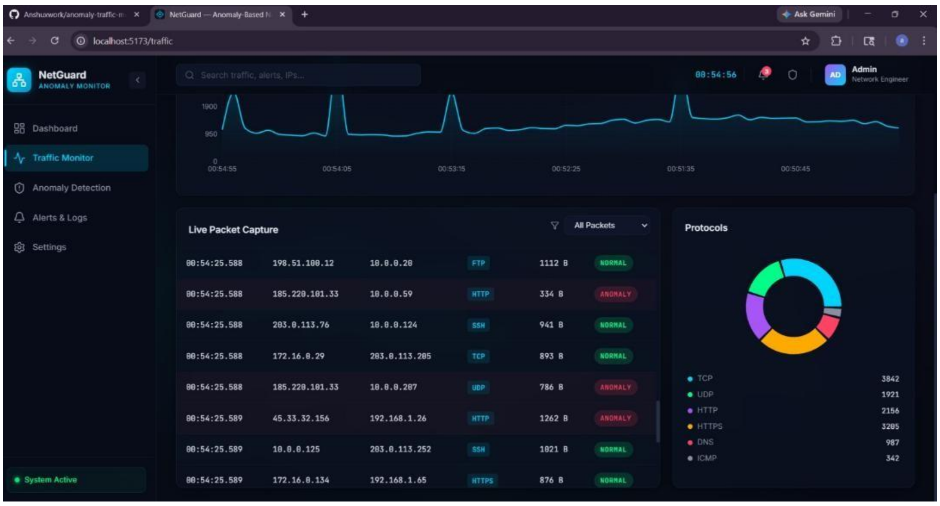
The protocol distribution chart shows the proportion of different network protocols such as TCP, UDP, HTTP, and HTTPS. This helps in understanding the nature of network traffic and identifying unusual protocol usage patterns that may signal suspicious activity.



4.3 Packet Capture and Analysis

The system captures live packets and displays them in a structured format including timestamp, source IP, destination IP, protocol, packet size, and status (Normal/Anomaly). This allows detailed inspection of network traffic in real time.

Some packets are flagged as “ANOMALY”, indicating deviations detected by the system. This demonstrates the effectiveness of the detection model in identifying suspicious activities within live network traffic



4.4 Anomaly Detection Results

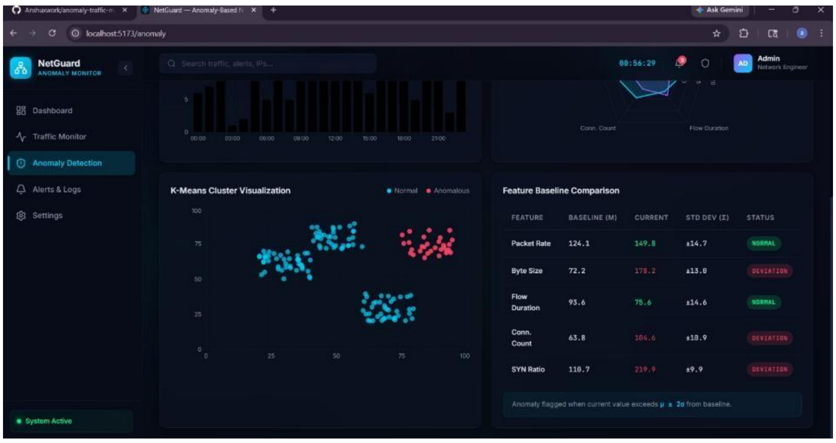
The anomaly detection module uses both Z-Score Analysis and K-Means Clustering (KMC). The clustering visualization clearly separates normal and anomalous data points, showing how the system identifies outliers.

The feature comparison table highlights deviations between baseline and current values for parameters such as packet rate, byte size, and connection count. When these values exceed the defined threshold, the system flags them as anomalies. This confirms that the system effectively detects abnormal behavior based on statistical deviation.

4.5 Alerts and System Performance

The system generates alerts whenever anomalies are detected. The dashboard displays the total number of anomalies and alerts in real time, allowing administrators to respond quickly. The alert system includes relevant details such as anomaly count and system status.

Overall, the system performs efficiently in real-time monitoring and detection. It successfully identifies abnormal traffic patterns with minimal delay and provides meaningful insights through visualization. The combination of statistical and clustering techniques improves detection accuracy while maintaining low computational overhead.



4.6 Discussion

The results demonstrate that the proposed system is capable of monitoring network traffic and detecting anomalies effectively. The visualization of traffic patterns and anomaly detection outputs makes the system user-friendly and practical for real-world applications.

While the system performs well in detecting major deviations, some limitations exist, such as possible false positives during sudden but legitimate traffic spikes. These can be reduced by fine-tuning threshold values and improving the baseline model. Overall, the system provides a balanced approach between accuracy, performance, and simplicity.

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CONCLUSIONS AND RECOMENDATIONS

5.1 Conclusion

This project successfully developed an Anomaly-Based Network Traffic Monitoring System (ABNTMS) capable of detecting abnormal patterns in network traffic using statistical and machine learning techniques. The system monitors real-time network activity, extracts relevant features, and identifies deviations from normal behavior using Z-Score analysis and K-Means Clustering (KMC). The results demonstrate that the system can effectively detect anomalies such as unusual traffic spikes and suspicious packet behavior.

The implementation of a real-time dashboard further enhances the usability of the system by providing clear visualization of traffic patterns, anomaly detection, and alert generation. The system achieves a balance between accuracy and performance while maintaining low computational complexity, making it suitable for academic and small-scale practical applications. Overall, the project highlights the effectiveness of anomaly-based approaches in improving network security compared to traditional signature-based methods.

5.2 Recommendations

Based on the results and observations, several improvements can be made to enhance the system:

The detection accuracy can be improved by integrating advanced machine learning or deep learning models such as LSTM or neural networks.

False positives can be reduced by fine-tuning threshold values and improving baseline modeling techniques.

The system can be extended to handle encrypted traffic by analyzing metadata such as packet size and timing.

Integration with automated response systems can help in taking immediate action against detected threats.

The dashboard can be further enhanced with more interactive visualizations and detailed analytics.

5.3 Future Scope

The project can be extended in several directions to improve its functionality and scalability. One possible enhancement is the implementation of deep learning-based anomaly detection models that can capture complex temporal patterns in network traffic. Additionally, deploying the system in a cloud or distributed environment can make it suitable for large-scale network infrastructures.

Further improvements may include integration with Security Information and Event Management (SIEM) systems, automated threat mitigation, and real-world dataset testing. The system can also be adapted for specific environments such as IoT networks or enterprise systems, where security requirements are more critical. These enhancements would make the system more robust, scalable, and suitable for real-world cybersecurity applications.

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